Low-Cost Amphibious Robotic System for Freshwater Wetland Research and Exploration

HILARIO ESPARZA, RYAN BRYNJOLFSON, AND KIRSTIN PETERSEN

Collective Embodied Intelligence Lab
www.ceiloce.com.edu

Motivation

Wetlands are large carbon sinks and produce 20-30% of the total annual global methane emissions. Understanding wetlands better will improve our understanding of the climate and the roles that this ecosystem plays within it.

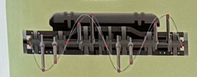
However, wetlands are notoriously difficult to navigate and traverse, posing as a barrier for research. New robotic platforms allow for the exploration of such ecosystems, but these tend to be prohibitively expensive and inaccessible to researchers

Our goal is to create an inexpensive, customizable amphibious robotic platform that would allow ecologists to map and monitor wetlands at greater spatiotemporal scales than conventional field measurements

Challenges

- **Amphibious Mobility:** The robot must be designed to have maneuverability and stable motion through rugged terrain, dense vegetation, and water.
- **Field Operation:** The robot will have to operate in a wide range of locations, where it is entirely possible that there will be no cellular service.
- **Waterproofing:** The robot must have an inner waterproof enclosure that protects the electronics from the environment, if not be entirely waterproof itself.
- **Accessibility:** To maintain a low barrier-of-entry, only general-purpose manufacturing equipment and 3D printers can be used to build and assemble the robot.

Motion



A single DC motor, per side, drives a traveling-wave motion through the robot's legs. This motion, coupled with the friction that the skin placed over the legs will create with the ground, will allow the robot to locomote on land and propel through water.

Waterproof Electronics Enclosure



A central enclosure houses the Raspberry Pi, DC motors and their controllers, LiPo batteries, RTK GNSS receiver, and two IMUs. Epoxy resin and rotary seals will waterproof the enclosure.

Visual Feedback



Two onboard cameras provide visual feedback for both the robot and the operator. A forward-facing camera can be used to observe the environment ahead of the robot, avoid obstacles, and detect species of flora and fauna.

The secondary camera provides additional visual information about the robot's state and its environment

Future work will use these cameras to perform autonomous obstacle avoidance and species recognition.



Skin



A waterproof, easy-to-assemble skin will wrap around the robot. This may be constructed from TPU or silicon, which will allow it to be flexible. This skin will create 'fins', enabling the to maneuver through water and increase friction when contacting locomoting on land. Additionally, the skin will further protect our robot from the harsh amphibious environments.

Sensors



Two orthogonally mounted IMUs can be used to estimate the robot's pitch, roll, and yaw.



A ZED-F9P-05B GNSS module is used to gather high accuracy outdoor positional data. A secondary receiver acts as a base station to provide correctional data.

Future work will increase the customizability of the sensor suite. This may include moisture sensors, pH sensors, and oxygen saturation sensors. Additionally, we plan to add an auger module, enabling soil data collection.

Approach

- We were inspired by the Velox platform developed by Pliant Robotics
- Off-the-shelf electronics were used to reduce cost and simplify replication
- We used a 3D-printed PLA chassis to increase iterability and rapid prototyping
- Separate rechargeable LiPo batteries were used for the motors and RPi 5, allowing for long term usage and reducing electrical noise.
- To improve the efficiency and speed of the robot when moving through water, we designed it to have a large length-to-width ratio
- A base station can communicate with the robot via a TCP socket connection, but radio communication may be used instead



Future Work

- Simplify chassis to reduce resource cost and improve ease of manufacturing
- Fabricate the skin and validate waterproofing
- Test stable locomotion in water and on land
- Integrate further ecological sensing capabilities
- Develop waypoint navigation and state-space control
- Introduce computer vision capabilities such as obstacle avoidance and species recognition

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CONTACT: Kirstin@cornell.edu